

Social Media Misinformation Detection

Module: Content-based Detection (NLP)

Literature Review

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1 INTRODUCTION AND SCOPE

Misinformation on social media rests on two coupled signals: the content of a post and the network that carries it. This review addresses the first. The task is to judge, from the text of a post, whether the claim it carries is reliable or misleading. It is one of two modules in the project; the companion module analyses propagation and account influence across the network. The two are designed to operate independently: this module produces a per-post credibility estimate, and passing that estimate to the network module is an integration step rather than a dependency.

Content-based detection has a substantial literature in natural language processing, which this review organises around the stages of a detection pipeline and the modelling choices within it. After framing the problem and its sub-tasks, we examine the available datasets, the modelling stack from classical features to pre-trained transformers, the central difficulty of generalisation, the question of explainability, the practical data challenges, and evaluation. A closing synthesis states the choices adopted for this module.

The analysis is conducted in English. The project additionally incorporates a small sample of real, collected posts, reserved as a realistic test set rather than used as primary training data.

2 PROBLEM FRAMING AND TASK DECOMPOSITION

The literature uses several overlapping terms. *Misinformation* denotes false or misleading content regardless of intent, *disinformation* implies deliberate deception, *rumour* refers to unverified claims in circulation, and *fake news* to fabricated articles imitating journalism (Shu et al., 2017; Zubiaga et al., 2018). This module targets the reliability of a post’s claim and sets intent aside, since intent is rarely observable from text alone.

Automated fact-checking is conventionally decomposed into a sequence of stages (Guo et al., 2022): identifying claims worth checking, gathering evidence and contextual signal, and issuing a verdict. This module follows that decomposition.

Check-worthiness. Most sentences are not factual claims that warrant verification. Claim-Buster casts claim spotting as supervised classification and scores sentences by how check-worthy they are (Hassan et al., 2017), which prevents later stages from expending effort on opinion and conversational filler.

Stance. The position that surrounding posts take toward a claim is an informative cue. RumourEval labels each reply in a thread as support, deny, query, or comment, alongside the rumour’s veracity (Derczynski et al., 2017), while the Fake News Challenge (FNC-1) frames the related problem of a news body’s stance toward its headline. Aggregated stance is a useful feature for the verdict.

Evidence retrieval. Verifying a claim against external sources reduces reliance on surface style. FEVER formalises this setting, pairing each claim with retrieved Wikipedia sentences and a label of *Supported*, *Refuted*, or *NotEnoughInfo* (Thorne et al., 2018).

Verdict. The final stage assigns a reliability label to the post and is the component on which most of this review concentrates. Considering the stages jointly, rather than the verdict in isolation, yields a system that is both more accurate and more interpretable.

Content-based analysis has an intrinsic ceiling. Some posts are misleading in ways the text does not reveal, through manipulated context, coordinated amplification, or claims that are literally true yet deceptively framed, and these are better exposed by how the content spreads than by what it states. The two-module design follows from this: the content verdict and the network’s propagation signal are complementary, and the stronger system combines them.

3 DATASETS

The dataset determines what a content model can learn and how its scores should be read. Table 1 lists the principal text-centric resources, mapped to the pipeline stage they serve.

Dataset	Size	Labels	Pipeline stage
LIAR	12.8K statements	6-way veracity	Verdict (text-only)
FakeNewsNet	News + engagement	Fake / real	Verdict (+ social context)
CoAID	COVID-19 news + tweets	Fake / real	Verdict (health)
FEVER	185K claims	Supported / Refuted / NEI	Check-worthiness, evidence
RumourEval	Twitter threads	Stance + veracity	Stance
Pérez-Rosas	7 news domains	Fake / legitimate	Verdict (linguistic study)

Table 1: Text-centric datasets, mapped to the pipeline stage they serve.

LIAR provides 12.8K labelled short statements from PolitiFact across six veracity levels (Wang, 2017); being text-only and balanced, it suits the verdict classifier, although its terse statements remove context and make the task harder than full articles. FakeNewsNet supplies news items with their social engagement (Shu et al., 2020); its text serves this module while its engagement graph serves the companion module. CoAID restricts the problem to COVID-19 health claims (Cui and Lee, 2020), a focused option for a single-domain study. FEVER recasts verification as checking claims against retrieved evidence (Thorne et al., 2018), connecting the check-worthiness and evidence stages. RumourEval contributes labelled stance and veracity over conversation threads (Derczynski et al., 2017). The seven-domain corpus of Pérez-Rosas et al. (2018) was constructed to expose linguistic differences between fake and legitimate news.

A consistent caveat across these resources is that a reported score is only as broad as the data behind it. A model trained and evaluated on a single topic and period can appear strong yet fail under a shift of topic, style, or time, which motivates holding the real, collected sample apart as a harder test rather than folding it into training.

4 MODELLING APPROACHES

4.1 FEATURE-BASED AND CLASSICAL MODELS

Before pre-trained language models, detection relied on hand-crafted features fed to a classical classifier such as logistic regression, a support vector machine, or a random forest. Castillo et al. (2011) assessed the credibility of Twitter content from the post text, from user posting and re-posting behaviour, and from citations to external sources. Pérez-Rosas et al. (2018) examined n-grams, punctuation, readability, and psycholinguistic categories, reaching accuracies around 76% across domains. These models are interpretable, since each feature carries an explicit meaning, and inexpensive to train; their limitation is brittleness, as cues that separate fake from real in one domain seldom transfer and can be evaded deliberately. They constitute a transparent baseline that a neural model should be required to surpass.

4.2 TRANSFORMER-BASED MODELS

Pre-trained transformers, fine-tuned on a labelled set, define the current state of practice. BERT introduced deep bidirectional pre-training with masked language modelling (Devlin et al., 2019); RoBERTa retained the architecture but improved the pre-training recipe and is generally the stronger encoder on classification (Liu et al., 2019); DeBERTa adds a disentangled content/position attention mechanism (He et al., 2021). In practice these models are accessed through the Hugging Face Transformers library, which reduces fine-tuning to a small amount of code (Wolf et al., 2020).

On misinformation benchmarks a fine-tuned RoBERTa is a strong baseline, typically matching or exceeding BERT, though such figures derive from narrow, single-topic test sets and should be read accordingly. Fine-tuning a base-size encoder is modest in compute and runs on a single GPU or a hosted notebook service. More recent prompting-based use of large language models offers an alternative without task-specific training, but fine-tuned encoders remain competitive on this task and are cheaper to serve (Guo et al., 2022). The plan adopted here uses RoBERTa as the main verdict model, BERT for comparison, and DeBERTa as an optional stronger encoder.

Fine-tuning in this regime is sensitive to a few practical choices. Once restricted to a single domain, misinformation datasets are often small, which makes a large encoder prone to overfitting; conservative learning rates, early stopping, and class weighting against imbalance are the usual remedies. Common practice clusters around a learning rate near 2×10^{-5} over a few epochs, and outcomes vary enough across random seeds that averaging several runs gives a more reliable estimate than a single one.

5 GENERALISATION AND ROBUSTNESS

The principal weakness of these models is generalisation. A score measured on a held-out split of the same dataset overstates real performance, because a model trained on one topic or period degrades when the topic, style, or period shifts. The implication is direct: misinformation evolves, and a detector that recognises only past narratives has limited value. The appropriate response is to measure the gap rather than conceal it, evaluating on cross-domain and cross-time splits and reporting the resulting drop alongside the in-domain figure.

Multilingual encoders such as XLM-R extend transformers across many languages from shared pre-training (Conneau et al., 2020). As the present analysis is in English, this lies outside the current scope, but it is the natural extension should other languages be required. Robustness also has an adversarial dimension, since stylistic detectors can be manipulated; grounding a verdict in retrieved evidence, as in FEVER (Thorne et al., 2018), reduces dependence on surface style.

6 EXPLAINABILITY

A credibility judgement is more useful, and more trustworthy, when accompanied by a justification. Two lines of work apply. Model-agnostic methods such as LIME explain any classifier by locally approximating it and identifying the tokens that drove a prediction (Ribeiro et al., 2016). Task-specific methods embed explanation in the model: dEFEND employs a sentence-comment co-attention network to surface the sentences and user comments most responsible for a verdict (Shu et al., 2019). For a transformer, attention weights and token attributions provide a first, low-cost layer of explanation.

Explainability carries particular weight in a security-sensitive setting, where an unexplained automated judgement is difficult to act upon. Surfacing the evidence behind a flag also assists development, by exposing cases in which the model relies on spurious correlations.

Explanations are themselves subject to evaluation. A highlighted span is useful only if it faithfully reflects the model’s reasoning rather than a plausible-looking rationalisation, and faithfulness need not coincide with how convincing an explanation appears to a reader. Because the explanations here serve both the analyst acting on a flag and the developer debugging the model, they are more valuable when checked against the model’s behaviour than when taken at face value.

7 DATA CHALLENGES

Beyond model choice, several data-level problems shape results. Misinformation datasets are typically imbalanced, with reliable content far outnumbering misleading content, so metrics and sampling must account for skew. Labels are noisy, since veracity annotation is difficult and annotators disagree, and acquiring labels at scale is costly. Models are prone to shortcut learning, latching onto dataset artefacts, such as source or topic markers, rather than the properties of misleading claims, which inflates in-domain scores and harms transfer. Content also drifts over time as narratives change, so a static test set ages.

These observations shape how the real, collected sample is used. Rather than enlarging the training set, it serves as an out-of-distribution test that reflects the conditions the system would face in practice, and as a check on whether in-domain gains survive a genuine shift.

Assembling that sample raises its own questions. Collection must respect platform terms of service and the privacy of the accounts involved, which constrains what may be stored and redistributed. Annotation needs a written protocol and more than one annotator per item so that agreement can be measured, since single-annotator labels conflate genuine signal with individual judgement. Because reliable content far outnumbers misleading content in an unfiltered stream, some targeted sampling is required to obtain a usable number of positive cases without distorting the test distribution beyond recognition.

8 EVALUATION METHODOLOGY

Because the datasets are imbalanced, plain accuracy is misleading; macro-averaged F1, which weights both classes equally, is the more informative headline metric, reported with precision and recall. ROC-AUC summarises ranking quality across thresholds, which is relevant because the companion module can consume a continuous credibility score rather than a hard label; for the same reason, score calibration is worth reporting. Beyond in-domain numbers, the evaluation should include at least one cross-domain or cross-time split to expose the generalisation gap. The check-worthiness and stance stages are assessed with the same ranking and macro-F1 measures, so the pipeline is evaluated end to end rather than at the verdict alone. Where latency matters, early detection, accuracy as a function of how much text is available, is a secondary axis worth

tracking.

Aggregate metrics should be read together with error analysis. The confusion matrix separates the two failure modes, missing misleading content and flagging reliable content, which carry different costs in deployment. Grouping errors by topic, length, and source often exposes systematic weaknesses, such as strong performance on political claims but weak performance on health, and that diagnosis guides the next iteration more usefully than a single aggregate figure.

9 DEPLOYMENT CONSIDERATIONS

The module is intended to run as a component, not only as an offline experiment, which raises a few practical points. Model size trades off against latency: a base encoder is accurate but heavier, whereas a distilled encoder runs faster at a modest cost in accuracy, which matters when many posts must be scored. Inference is naturally batched, and throughput rather than single-example speed is usually the binding constraint. The operating point must also be chosen deliberately, since turning a continuous score into a flag requires a threshold whose value depends on the relative cost of the two error types and on what the network module expects as input. Finally, because content drifts, the verdict model needs periodic re-training on fresh data, with the evaluation splits refreshed alongside it.

10 SYNTHESIS: CHOICES FOR THIS MODULE

The literature supports a staged plan, summarised in Table 2.

Stage	Approach	Data / metric
Check-worthiness	Rank sentences by claim worth (ClaimBuster-style)	Labelled claims; ranking quality
Stance	Classify replies / bodies (support, deny, ...)	RumourEval, FNC-1; macro-F1
Verdict (main)	Fine-tune RoBERTa; BERT baseline, DeBERTa optional	LIAR then FakeNewsNet; macro-F1, ROC-AUC
Generalisation	Cross-domain / cross-time splits, report the drop	Held-out domain; macro-F1 gap
Explainability	Token attribution / attention; DEFEND-style if comments exist	Qualitative, check-worthy spans

Table 2: Recommended pipeline for the content-based module.

In summary, the module is built as a staged pipeline rather than a single classifier. An interpretable feature-based baseline precedes a fine-tuned RoBERTa verdict model, each stage is assessed with macro-F1, and ROC-AUC with the verdict, under both in-domain and shifted splits. Each prediction carries an explanation, and a small real, collected sample is retained as an out-of-distribution English test set. The module emits a per-post credibility score that the network module may consume, while each module remains independently runnable.

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